PROBLEM 1.6

Saline Mixing Tank Dynamics (CSTR)

Original statement · ProblemsCh1.pdf, p. 2

CSTR Mass Balance • Washout • Integrating Factor

PHASE 1: PROBLEM STATEMENT & HYPOTHESES

Problem statement: A tank contains 100 liters of water with 1 kg of salt dissolved in it (uniformly). At time $t_0 = 0$, water without salt flows into the tank at a rate of 1 liter per minute, while the mixture flows out at the same rate.

(i) Write a differential equation for $x(t)$, which is the number of kg of salt per litre of water dissolved at time $t > 0$.
(ii) How much salt is dissolved in the tank as $t \to \infty$?
(iii) What happens if water with s kg of salt per liter flows into the tank at a given rate while the mixture (that is kept uniform) flows out at the same rate?

PHASE 2: FUNDAMENTAL FORMULATION

As in the official statement, $x(t)$ is the salt **concentration** (kg of salt per litre, units $\mathrm{kg/L}$). The total dissolved salt mass is $M(t) = V(t)\,x(t)$ (units kg). Species conservation of dissolved salt mass: $\frac{dM}{dt} = \dot{m}_{\text{in}} - \dot{m}_{\text{out}} = Q_{\text{in}} C_{\text{in}} - Q_{\text{out}}\,x(t)$. Since $Q_{\text{in}} = Q_{\text{out}} = Q$, $V(t) \equiv V_0 = 100\ \mathrm{L}$, so $M(t) = 100\,x(t)$ and the initial concentration is $x(0) = \frac{1\ \mathrm{kg}}{100\ \mathrm{L}} = 0.01\ \mathrm{kg/L}$.

PHASE 3: STEP-BY-STEP DERIVATION

Part (i): Pure Water Flushing ($C_{\text{in}} = 0$):

$$\dot{m}_{\text{in}} = 1 \times 0 = 0\ \mathrm{kg/min}, \quad \dot{m}_{\text{out}} = 1 \times x(t) = x(t)\ \mathrm{kg/min}$$

With $M = 100\,x$, the balance $\frac{dM}{dt} = \dot{m}_{\text{in}} - \dot{m}_{\text{out}}$ becomes $100\,\frac{dx}{dt} = -x(t)$, i.e.

$$\boxed{\frac{\mathbf{dx}}{\mathbf{dt}} = -\frac{\mathbf{x(t)}}{\mathbf{100}}, \quad x(0) = 0.01 \text{ kg/L}} \tag{1.6.1}$$

Part (ii): Analytical Solution and Limit as $t \to \infty$:

Equation (1.6.1) is the population equation of the class notes with constant $-\frac{1}{100}$ [Notes, p. 2]. Since $x > 0$, divide by x and integrate from 0 to t; on the left substitute $u = x(\tau)$, $du = x'(\tau)\,d\tau$, with limits $u = 0.01$ and $u = x(t)$:

$$\int_0^t \frac{x'(\tau)}{x(\tau)}\,d\tau = \int_{0.01}^{x(t)} \frac{du}{u} = \int_0^t -\frac{1}{100}\,d\tau \implies \ln\frac{x(t)}{0.01} = -\frac{t}{100} \implies \boxed{\mathbf{x(t) = 0.01\,e^{-t/100}}\ \text{kg/L}}$$

The dissolved salt mass is $M(t) = 100\,x(t) = e^{-t/100}$ kg. Taking the limit as $t \to \infty$:

$$\lim_{t\to\infty} x(t) = \mathbf{0}\ \text{kg/L}, \qquad \lim_{t\to\infty} M(t) = \lim_{t\to\infty} e^{-t/100} = \mathbf{0}\ \text{kg}$$

Part (iii): Saline Inflow at Concentration s kg/L:

Now $\dot m_{\text{in}} = Q \cdot s = 1 \cdot s = s$ kg/min, so $100\,\frac{dx}{dt} = s - x(t)$, leading to the inhomogeneous linear ODE:

$$\boxed{\frac{dx}{dt} + \frac{1}{100}x(t) = \frac{s}{100}} \tag{1.6.2}$$

Three-step algorithm of the class notes [Notes, p. 3], with $p = \frac{1}{100}$, $q = \frac{s}{100}$, $t_0 = 0$, $x(0) = 0.01$:

1. **Find I.** $P(t) = \int_0^t \frac{d\tau}{100} = \frac{t}{100}$, so $I(t) = e^{t/100}$ and $I(0) = 1$.
2. **Integrate from 0 to t.** $\int_0^t \left(e^{\tau/100}x(\tau)\right)'\,d\tau = \int_0^t \frac{s}{100}e^{\tau/100}\,d\tau$. On the right substitute $u = \frac{\tau}{100}$, $du = \frac{d\tau}{100}$: $s\int_0^{t/100} e^u\,du = s\left(e^{t/100} - 1\right)$.
3. **Simplify.** Barrow's rule on the left: $x(t)e^{t/100} - 1\cdot 0.01 = s\left(e^{t/100} - 1\right)$.

$$x(t)e^{t/100} = 0.01 + se^{t/100} - s = se^{t/100} + (0.01 - s)$$

Dividing by $e^{t/100}$:

$$\boxed{\mathbf{x(t) = s + (0.01 - s)e^{-t/100}}\ \text{kg/L}} \tag{1.6.3}$$

The corresponding dissolved salt mass is $M(t) = 100\,x(t) = 100s + (1 - 100s)e^{-t/100}$ kg. As $t \to \infty$, $e^{-t/100} \to 0$:

$$\lim_{t\to\infty} x(t) = \mathbf{s}\ \text{kg/L}, \qquad \lim_{t\to\infty} M(t) = \mathbf{100s}\ \text{kg}$$

PHASE 4: FINAL RESULT & ANALYSIS

BOXED RESULT

$$\text{Washout: } x(t) = 0.01\, e^{-t/100}\ \text{kg/L},\ M(t) = e^{-t/100}\ \text{kg},\ \lim_{t\to\infty} M(t) = 0$$

$$\text{Saline Inflow: } x(t) = s + (0.01 - s)e^{-t/100}\ \text{kg/L},\ \lim_{t\to\infty} x(t) = s\ \text{kg/L},\ \lim_{t\to\infty} M(t) = 100s\ \text{kg}$$

Physical Interpretation: The asymptotic equilibrium concentration is the inlet concentration s kg/L, i.e. a dissolved mass $100s$ kg (tank volume 100 L times s kg/L).